



LEADERS IN ENVIRONMENTAL MONITORING



SCIENTIFIC SANITATION SOLUTIONS

**Water Quality Report
81256**

February 2020

**Compiled by: Aquatico Scientific
(012 450 3800)**

ATTENTION: SCIENTIFIC SANITATION SOLUTIONS (PTY) LTD

17 FEBRUARY 2020

DRINKING WATER REPORT BASED ON TEST REPORT 81256

INTRODUCTION:

Aquatico was commissioned by Scientific Sanitation Solutions (Pty) Ltd to analyse and evaluate the physical, chemical and bacteriological water quality of a single sample (**Sani Amanzi**). The water being evaluated was subjected to a purification product (added before and after where the sample was taken) to improve quality.

This evaluation will aim at determining the efficacy of the purification process in terms of a domestic use standard and if the submitted water sample is fit for domestic use. The sample was submitted to Aquatico Laboratory for analysis on 27 January 2020.

It is important to note that the product that was administered is unknown to Aquatico and therefore this report will solely focus on quality attained based on the selected array of variables, as per client's request.

Water quality is compared against the SANS 241-1:2015 drinking water standards.

Water quality is classified according to the Quality of Domestic water supplies (WRC, 1998) standard classification system (See Table 1). When comparing the data to the guidelines; the worst substance class will determine the overall class of the water supply. Where there exists no limit in the WRC Guideline for a particular variable, the DWAF (1996) Domestic Use Guideline will be implemented.

Table 1: WRC Quality of Domestic Water Supplies – Colour classification system.

Class / Colour	Description	Effects
Class 0	Ideal water quality	No effects, suitable for many generations
Class 1	Good water quality	Suitable for lifetime use. Rare instances of sub-clinical effects
Class 2	Marginal water quality	May cause some effects in sensitive users. Some effects possible after a lifetime of use. Aesthetic effects.
Class 3	Poor water quality	Poses risk of chronic health effects, especially in babies, children and the elderly. Poor aesthetics
Class 4	Unacceptable water quality	Severe acute health effects, even with short-term use. Taste and appearance will lead to rejection of the water.



RESULTS:

Table 2: Water quality results from the representative sample for January 2020.

VARIABLE	UNITS	SANS 241-1:2015 Drinking Water Standard (SABS, 2015)	Quality of Domestic Water Supplies: Drinking Class 1 Upper range	MONITORING LOCALITIES
				Sani Amanzi
pH @ 25°C	pH	5.0/9.7	4.5/10.0	7.55
Electrical conductivity (EC) @ 25°C	mS/m	170	150	40.3
Total Dissolved solids @ 180°C	mg/l	1200	1000	260
Chloride (Cl)	mg/l	300	200	13.7
Sulphate (SO ₄)	mg/l	500	400	61.9
Nitrate (NO ₃) as N	mg/l	11	10	1.1
Nitrite (NO ₂) as N	mg/l	0.9	-	<0.065
Ammonium (NH ₄) as N	mg/l	1.5	-	0.789
Ammonia (NH ₃) as N	mg/l	-	-	0.014
Fluoride (F)	mg/l	1.5	1	<0.263
E.coli	CFU/100ml	0	1	0
Total coliform	CFU/100ml	10	10	0
TotalViableCount	CFU/ml	1000	-	4
Turbidity	NTU	1	1	0.95
Free chlorine (Cl ₂)	mg/l	5	0.6	0.36
Color	mg/l Pt-Co	15	-	7
Phenol	mg/l	0.01	-	<0.010
Total organic carbon (TOC)	mg/l	10	-	6.55
Temperature	°C	-	-	22.1
Trihalomethanes (THM)	µg/l	-	-	41
Bromoform	µg/l	100	-	<2
Chloroform	µg/l	300	-	30
Dibromochloromethane	µg/l	100	-	2
Bromodichloromethane	µg/l	60	-	9
Ammonia (NH ₃) as NH ₃	mg/l	1.5	-	0.017
Microcystin	µg/l	1	-	<0.500
Monochloramine	mg/l	3	-	0.39
Nitrate + Nitrite as N	mg/l	-	-	5
Total Cyanide (CN)	mg/l	0.2	-	<0.005
Acid Soluble Sodium (Na)	mg/l	-	-	24.7
Acid Soluble Aluminium (Al)	mg/l	0.3	-	0.024
Acid Soluble Cadmium (Cd)	mg/l	0.003	-	<0.002
Acid Soluble Chromium (Cr)	mg/l	0.05	-	<0.003
Acid Soluble Copper (Cu)	mg/l	2	-	0.017
Acid Soluble Iron (Fe)	mg/l	0.3	-	0.078
Acid Soluble Manganese (Mn)	mg/l	0.1	-	0.004
Acid Soluble Nickel (Ni)	mg/l	0.07	-	0.01
Acid Soluble Lead (Pb)	mg/l	0.01	-	<0.004
Acid Soluble Zinc (Zn)	mg/l	5	-	0.06
Acid Soluble Boron (B)	mg/l	2.4	-	<0.013
Acid Soluble Barium (Ba)	mg/l	0.7	-	0.051
Acid Soluble Antimony (Sb)	mg/l	-	-	<0.001
Acid Soluble Arsenic (As)	mg/l	0.01	-	<0.006
Acid Soluble Selenium (Se)	mg/l	0.04	-	<0.002
Acid Soluble Mercury (Hg)	mg/l	0.006	-	<0.004
Acid Soluble Uranium (U)	mg/l	0.03	-	<0.015

- The physical characteristic of **Sani Amanzi** can be described as neutral (pH 6.0 – 8.5), non-saline (TDS < 450 mg/l) and moderately soft (Total Hardness 50 - 100 mg CaCO₃/l) with no *E.coli* and no Total Coliforms detected (Table 2).
- Based on the presented variables in Table 2 it can be seen that the submitted water quality complied with the SANS 241-1:2015 limits in terms of all the analysed variables.
- According to the Quality of Domestic water supplies (WRC, 1998) standard classification, the water quality of the submitted sample can be classified as **Good (Class 1)** water quality.

DISCUSSION:

The physical characteristic of the water sample can be described as neutral (pH 6.0 – 8.5), non-saline (TDS < 450 mg/l) and moderately soft (Total Hardness 50 - 100 mg CaCO₃/l). No *E.coli* and no Total Coliforms were detected.

None of the analysed variables exceeded the SANS 241-1:2015 Drinking Water Standard. Furthermore, this water can be classified as Good (Class 01) water quality and is fit for use as potable water and domestic purposes.

CONCLUSION:

One water sample, named **Sani Amanzi**, was evaluated according to its physical, chemical and bacteriological water quality to determine the water is fit for use as potable water and domestic purposes after a purification product was added.

All analysed variables complied with the SANS 241-1:2015 drinking water standard limits and the water can be classified as Good (Class 01) water quality.

The water is fit for use as potable water and domestic purposes.

REFERENCES:

SABS [South African Bureau for Standards] SANS 241-1:2015 South African National Standard for Drinking Water Part 1: Microbiological, physical, aesthetic and chemical determinants. SABS Standards Division. Pretoria.

WRC [Water Research Commission], DWAF [Department of Water Affairs and Forestry], and DOH [Department of Health] (1998). Quality of Domestic Water Supplies. Volume 1: Assessment Guide. (2nd Edition). WRC Report no. TT 101/98. Pretoria.

Appendix A

TEST REPORT

Test Report

Page 1 of 2

Client: Scientific Sanitation Solutions Pty Ltd
Address: 323 Lynnwood Road, Menlo Park, 0047
Report no: 81256
Project: Scientific Sanitation Solutions Pty Ltd

Date of certificate: 13 February 2020
Date accepted: 28 January 2020
Date completed: 13 February 2020
Date received: 27 January 2020

Lab no:			81562
Date sampled:			27-Jan-20
Aquatico sampled:			No
Sample type:			Water
Locality description:			Sani Amanzi
Analyses			
	Unit	Method	
A pH @ 25°C	pH	ALM 20	7.55
A Electrical conductivity (EC) @ 25°C	mS/m	ALM 20	40.3
A Total Dissolved solids @ 180°C	mg/l	ALM 24	260
A Chloride (Cl)	mg/l	ALM 02	13.7
A Sulphate (SO ₄)	mg/l	ALM 03	61.9
A Nitrate (NO ₃) as N	mg/l	ALM 06	1.10
A Nitrite (NO ₂) as N	mg/l	ALM 07	<0.065
A Ammonium (NH ₄) as N	mg/l	ALM 05	0.789
N Ammonia (NH ₃) as N	mg/l	ALM 26	0.014
A Fluoride (F)	mg/l	ALM 08	<0.263
A Acid Soluble Sodium (Na)	mg/l	ALM 30	24.7
A Acid Soluble Aluminium (Al)	mg/l	ALM 31	0.024
A Acid Soluble Iron (Fe)	mg/l	ALM 31	0.078
A Acid Soluble Manganese (Mn)	mg/l	ALM 31	0.004
A Acid Soluble Chromium (Cr)	mg/l	ALM 31	<0.003
A Acid Soluble Copper (Cu)	mg/l	ALM 31	0.017
A Acid Soluble Nickel (Ni)	mg/l	ALM 31	0.010
A Acid Soluble Zinc (Zn)	mg/l	ALM 31	0.060
A Acid Soluble Cadmium (Cd)	mg/l	ALM 31	<0.002
A Acid Soluble Lead (Pb)	mg/l	ALM 31	<0.004
A E.coli	CFU/100ml	ALM 40	<1
A Total coliform	CFU/100ml	ALM 40	<1
A TotalViableCount	CFU/ml	ALM 43	4
A Turbidity	NTU	ALM 21	0.950
N Free chlorine (Cl ₂)	mg/l	ALM 23	0.36
N Color	mg/l Pt-Co	ALM 22	7.00
A Total Cyanide (CN)	mg/l	ALM 16	<0.005
N Phenol	mg/l	ALM 71	<0.010

A = Accredited N = Non accredited Out = Outsourced Sub = Sub-contracted NR = Not requested RTF = Results to follow NATD = Not able to determine ATR = Alternative test report ; The results relates only to the test item tested; Results reported against the limit of detection; Results marked 'Non SANAS Accredited' in this report are not included in the SANAS Schedule of Accreditation for this laboratory; Uncertainty of measurement available on request for all methods included in the SANAS Schedule of Accreditation; The report shall not be reproduced except in full without approval of the laboratory
 The results apply to the sample received.

Test Report

Page 2 of 2

Client: Scientific Sanitation Solutions Pty Ltd
Address: 323 Lynnwood Road, Menlo Park, 0047
Report no: 81256
Project: Scientific Sanitation Solutions Pty Ltd

Date of certificate: 13 February 2020
Date accepted: 28 January 2020
Date completed: 13 February 2020
Date received: 27 January 2020

Lab no:		81562		
Date sampled:		27-Jan-20		
Aquatico sampled:		No		
Sample type:		Water		
Locality description:		Sani Amanzi		
Analyses		Unit	Method	
A	Total organic carbon (TOC)	mg/l	ALM 63	6.55
A	Acid Soluble Arsenic (As)	mg/l	ALM 34	<0.006
A	Acid Soluble Selenium (Se)	mg/l	ALM 34	<0.002
N	Acid Soluble Mercury (Hg)	mg/l	ALM 35	<0.004
A	Acid Soluble Boron (B)	mg/l	ALM 33	<0.013
A	Acid Soluble Barium (Ba)	mg/l	ALM 33	0.051
A	Acid Soluble Uranium (U)	mg/l	ALM 37	<0.015
N	Acid Soluble Antimony (Sb)	mg/l	ALM 36	<0.001
A	Nitrate + Nitrite as N	mg/l	ALM 26	5.00
N	Ammonia (NH ₃) as NH ₃	mg/l	ALM 26	0.017
N	Temperature	°C	ALM 20	22.1
N	Trihalomethanes (THM)	µg/l	OLM 02	41
N	Bromoform	µg/l	OLM 02	<2
N	Chloroform	µg/l	OLM 02	30
N	Dibromochloromethane	µg/l	OLM 02	2
N	Bromodichloromethane	µg/l	OLM 02	9
N	Monochloramine	mg/l	ALM 67	0.39
N	Microcystin	µg/l	ALM 68	<0.500

A = Accredited N = Non accredited Out = Outsourced Sub = Sub-contracted NR = Not requested RTF = Results to follow NATD = Not able to determine ATR = Alternative test report ; The results relates only to the test item tested; Results reported against the limit of detection; Results marked 'Non SANAS Accredited' in this report are not included in the SANAS Schedule of Accreditation for this laboratory; Uncertainty of measurement available on request for all methods included in the SANAS Schedule of Accreditation; The report shall not be reproduced except in full without approval of the laboratory
 The results apply to the sample received.

Authenticated signature on first page